

Ashok Kumar Parihar

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SUMMARY

- Data Analytics professional with 2+ years of experience at Tata Consultancy Services, specializing in IT infrastructure, automation, and data-driven decision-making. Proficient in SQL, Python, PowerShell, and machine learning, with strong knowledge of cloud platforms and virtualization tools. Currently completing a Master's in Applied Data Analytics at the University of Detroit Mercy, with hands-on experience in academic research, predictive modeling, and real-world analytics projects. Actively seeking full-time opportunities in data analytics, business intelligence, or data science roles.

EDUCATION

- **University of Detroit Mercy**, Detroit, MI
Master of Science in Applied Data Analytics
Graduated: May 2025 | CGPA: 3.5
- **Anurag University**, Hyderabad, India
Bachelor of Technology in Mechanical engineering
Graduated: 2021 | CGPA: 7.98

PROFESSIONAL EXPERIENCE

Tata Consultancy Services (TCS)

Hyderabad, India

System Engineer

November 2021 – December 2023

Responsibilities:

- Developed and automated performance reports using PowerShell scripting to analyze server health metrics, enabling data-driven decision-making and improving system reliability.
- Utilized SQL to query and analyze large datasets from IT infrastructure systems, identifying trends and optimizing resource utilization across servers and virtual machines.
- Conducted data analysis on incident logs and infrastructure performance, presenting actionable insights to reduce system downtime and streamline operations.
- Created dashboards and visual reports to monitor key performance indicators (KPIs) for server environments, facilitating proactive management and reporting.
- Migrated systems and optimized virtualized environments using VMware and SCCM, ensuring efficient resource allocation and improved infrastructure performance.
- Collaborated with cross-functional teams to analyze process gaps and implement automation solutions, increasing operational efficiency by reducing manual effort.
- Generated compliance reports and audited server activities, ensuring alignment with organizational standards and security protocols.

University of Detroit mercy

Detroit, United states of America

Graduate Assistant

Winter 2024

Responsibilities:

- Conducted **data analysis** and **report writing** for a research project on the **Boys and Girls Club**, improving understanding of community impact.
- Assisted in designing research methodologies and performed data collection to ensure accurate reporting and actionable outcomes.

University of Detroit mercy

Detroit, United states of America

Aeroponics Project Manager

Winter 2024

Responsibilities:

- *Managed data monitoring for plant health and nutrient systems in the aeroponics project, analyzing performance trends and documenting results to support future research.*
- *Utilized Excel and other reporting tools to measure project metrics and streamline documentation for progress analysis.*

University of Detroit mercy

Detroit, United states of America

Audio Assistant

Fall 2024

- Assisted professors with managing and organizing podcast **metadata** and uploading content to the university website, ensuring accurate tracking and easy access.
- Set up microphones and recording equipment for podcasts and university events, ensuring high-quality audio production.
- Developed and maintained a **tracking system** for podcast schedules and performance data, providing insights into content reach and engagement.
- Analyzed **listener data** to identify trends and improve future content strategies.
- Streamlined workflows by automating routine tasks, improving efficiency in content management.

University of Detroit mercy

Detroit, United states of America

Research Assistant – Vehicle Cyber Engineering (VCE) Lab

Spring 2024

- Developed Python scripts to automate the conversion of CAN log files into CSV format, enhancing data preprocessing for analysis workflows.
- Performed anomaly detection and CAN ID classification using Isolation Forest and Random Forest models, achieving over 99% accuracy.
- Connected and configured the Woodpecker CAN device with BusMaster for real-time message logging and diagnostics.
- Built a secure Python-based MQTT bridge to transmit CAN data with cloud certificates, enabling remote monitoring of vehicle telemetry.
- Reviewed research papers on ECU fingerprinting, anomaly detection, and automotive digital forensics; documented findings in the VCE lab notebook.
- Contributed weekly project status updates and shared scripts via the official VCE GitHub repository.

SKILLS

- **Programming Languages:** Python, SQL, R, PowerShell.
- **Big Data Technologies:** Apache Spark, Hadoop.
- **Data Warehousing:** Snowflake, Amazon Redshift, PostgreSQL, MySQL, Oracle, MongoDB.
- **Data Visualization:** Power BI, Tableau, Excel.
- **Databases (SQL/NoSQL):** Oracle, MySQL, Snowflake, Redshift, PostgreSQL, MongoDB.
- **Statistical tools:** STATA, SPSS, Data cleaning & ETL processes.
- **AI & Machine Learning:** Experience with ML frameworks including PyTorch, Scikit-learn, and TensorFlow, Machine Learning algorithms.
- **Cloud & Virtualization:** VMware, ESXi, vCenter, vSAN, AWS.
- **Other Tools:** Git, Jupyter Notebooks, SCCM.

PROJECTS

Amway-Alticor Infrastructure Support

Tata Consultancy Services (TCS), Hyderabad, India

November 2021 – December 2023

- Brief Description: Supported Amway-Alticor's IT infrastructure, ensuring high availability and performance across critical systems.
- Role/Responsibilities:
 1. Managed Windows Server, Active Directory, VMware, and SCCM.
 2. Performed server migrations and upgrades, optimizing system performance.
 3. Automated tasks with PowerShell and maintained system health.
- Technologies Used:

Windows Server, VMware vCenter, SCCM, PowerShell
- Impact:
 1. Migrated 500+ servers, reducing downtime by 20%, Upgraded 500+ servers to 2019 Operating system from 2008/2012 Operating system.
 2. Automated 25% of tasks, improving efficiency.

Chatbot Development Project

University of Detroit Mercy (January 2024 – April 2024)

- Description: Developed a Python-based chatbot using advanced NLP techniques (Lemmatization with Lemma1 and Lemma2) to streamline the University of Detroit Mercy's course registration process.
- Role: Developer
- Key Tasks:
 1. Designed and implemented a chatbot with natural language processing.
 2. Integrated features to enhance user interaction and automate course registration.
- Impact: Increased successful registrations by 25% in the first month and reduced manual workload.

Air Cargo Management System (August 2024 – December 2024)

University of Detroit Mercy

- Description: Designed a relational database to streamline air cargo logistics, enabling real-time shipment tracking, route optimization, and improved operational efficiency.
- Role: Database Designer
- Key Tasks:
 1. Developed a normalized database schema in MySQL.
 2. Created SQL queries for tracking and reporting.
 3. Ensured data integrity and scalability.
- Impact: Reduced delays, enhanced transparency, and improved logistics efficiency.

Lung Cancer Prediction Using Machine Learning (August 2024 – December 2024)

University of Detroit Mercy

- Description: Developed a machine learning model to predict lung cancer risk using patient data, aiming to improve early detection and intervention.
- Role/Responsibilities: Data preprocessing, feature selection, model training, performance evaluation, and report documentation.
- Key Tasks: Cleaned data, selected significant features, trained Logistic Regression, Decision Tree, and Random Forest models, evaluated performance using accuracy and F1-score, and visualized findings.
- Technologies Used: Python (pandas, scikit-learn, matplotlib), Kaggle, Google Collab
- Results/Impact: Achieved 97.5% accuracy with Random Forest, identified key risk factors for clinical use, and delivered a reliable pipeline for healthcare predictions.

CERTIFICATIONS

- Social and Behavioural Responsible Conduct of Research Course 1.

ACHIEVEMENTS

First Prize – Detroit Urban Health AI Challenge (Chatbot AI Track)

Collaborated on designing and developing an *Insomnia Personal Assistant Chatbot* using Generative AI on the Scholar AID platform. Contributed by creating and training datasets, refining bot responses, and enhancing its functionality to assist patients with insomnia management. Strengthened skills in AI application, teamwork, and problem-solving under time constraints.